MIL-A-37819A
18 August 1978
SUPERSEDING
MIL-A-37819 (DLA-DM)
13 July 1977

MILITARY SPECIFICATION

ARTICULATOR, DENTAL, PLAIN LINE, LINGUAL ACCESS DESIGN, WITH CONDYLAR AND AUDITORY PINS, AND REPLACEMENT MOUNTING INSERTS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an articulator of lingual access design with condylar and auditory pins and accessory mounting inserts, for use in mounting dental prosthesis.

1.2 Classification. The articulator and replacement mounting inserts shall be of the following types, as specified (see 6.1):

Type I. - Articulator, complete.
Type II. - Replacement mounting inserts.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-601 - Aluminum-alloy Sand Castings.
QQ-N-290 - Nickel Plating (Electrodeposited).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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MIL-P-116 - Preservation-Packaging, Methods Of.
MIL-S-46049 - Steel, Carbon, Strip, Cold Rolled, Hardened And Tempered Spring Quality.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWING

DEFENSE PERSONNEL SUPPORT CENTER (DPSC)

24391 - Articulator, Dental, Plain Line, Lingual Access Design.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 18 - Tests for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

ASTM E 92 - Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

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3. REQUIREMENTS

3.1 Material. All materials used shall be of high commercial quality throughout and shall be entirely suitable for the purpose intended. The materials shall be as specified in table I for the components and quantities listed.

TABLE I. Components, materials and finishes.

Quantity	Component	Material	Plating or coating	Finish
1 1	Upper bow Lower bow	Cast Aluminum Alloy No. 356 in accordance with QQ-A-601 for composition only.	Hard coat anodized	Tumbled
1	Spring	Spring Steel No. C1075 or C1095 in accordance with MIL-S-46049 for composition only. Rockwell hardness 30N 66-68.5	Nickel plated (0.0004 inch minimum)	-
1 1 2	Thumbscrew Thumbnut Mounting thumbscrew	Aluminum Alloy No. 2011-T3	- - Clear anodized	Knurled grasping edges Smooth
2	Condylar pin	Brass, yellow, free cutting	-	Machine
2	Screw (binding head)	Brass	Nickel	-
2	Nut	Brass, yellow, free cutting	-	Machine
2	Auditory pin	Steel, high carbon	-	Black
See 3.3.1 and 3.3.2	Mounting inserts	Aluminum Alloy No. 2011-T3	-	Machine

3.2 Hardness. The steel spring shall have a hardness as specified in table I when tested in accordance with 4.2.4.3.

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3.3 Style, design, and dimensions. Type I complete articulator shall be suitable for use within the dental prosthetic laboratory for occluding maxillary and mandibular anatomical casts. Type II replacement mounting inserts shall be suitable for use as replacement inserts for Type I articulator. The style, design, and dimensions of the complete articulator (3.3.1) and replacement mounting inserts (3.3.2) shall be in accordance with Defense Personnel Support Center (DPSC) Drawing 24391.

3.3.1 Type I, Articulator, complete. The articulator shall be furnished complete with 14 mounting inserts. Two inserts shall be mounted on the articulator (one each to the upper and lower bow) and twelve shall be furnished in a separate package. The articulator shall have a lower and upper bow. The lower bow shall be provided with two (right and left) condylar elements and two (right and left) auditory pins. The lower bow shall be provided with a thumbnut and thumbscrew for vertical height adjustment of the upper bow (maxillary cast portion). The upper bow shall be provided with a spring which shall retain the upper bow upon the lower bow condylar elements, permit protrusion movement and lateral excursions of the upper bow, return the upper bow to the centric when the upper bow is released from any eccentric position, and permit the removal of the upper bow without removal of the spring. The mounting inserts shall be suitable for mounting casts to the upper and lower bows of Type I articulator. The inserts shall be disposable, replaceable, and shall be attachable to the upper and lower bows by thumbscrews. The inserts shall permit the removal and precise replacement of casts. The upper and lower bows shall have circular male bosses which form registration keys on the cast. The articulator's condylar and auditory pins shall be suitable for use with NSN 6520-00-935-5826, FACE BOW, ARTICULATOR, DENTAL.

3.3.2 Type II, Replacement mounting inserts. The replacement mounting inserts shall be furnished in a package which shall contain 12 inserts. The replacement inserts shall meet all requirements of the specification applicable to the mounting inserts.

3.4 Drawings. The drawings forming a part of this specification are specification control drawings. The supplier shall be responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in improper design, the supplier shall provide a product with dimensions within the tolerances prescribed on the drawing to insure proper design, function, fit, assembly, and operation of the item. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.5 Weight. The weight of the articulator shall be 16 $\pm$ 1 ounces.

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3.6 Finish. The finish of the components shall be as specified in table I. Components shall be free of burrs and sharp edges. Plating shall be adherent, uniform in appearance, free from visible porosity, blisters, or other defects that may cause peeling. The anodic coating shall be uniform in appearance, clean, adherent, free from breaks, scratches, and other defects which reduce the serviceability of the anodized parts.

3.7 Identification marking. The articulator shall be marked with the name or registered trademark of the manufacturer. Articulators of foreign (other than U.S.) manufacture shall be marked with the name of the country of origin, together with the name or registered trademark of the contractor. All marking shall be permanent and legible.

3.8 Descriptive literature. Each articulator shall be furnished with literature containing articulator instructions together with manufacturer's part numbers for all components.

3.9 Spare parts. Each articulator shall be furnished with one spare binding head screw (a total of two screws for each articulator).

3.10 Workmanship. Workmanship shall be first class throughout. The casting shall be free from cracks and other defects (such as gas holes) which, due to their nature, degree, or extent, detract from its appearance or detrimentally affect its serviceability, proper functioning, or intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component or material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and III (see 4.2.1 and 4.2.2) and table IV (see 4.2.3).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and the acceptable quality levels (AQL's) shall be as indicated in table II. The unit of product for examination purposes shall be one complete articulator (including mounting inserts) or one package of replacement mounting inserts, as applicable.

TABLE II. Examination criteria.

Type of defect	Inspection level	AQL (percent defective)
Major A	I	1.0
Major B	S-2	2.5
Minor	I	4.0

4.2.2 Examination. The complete articulator (including mounting inserts) or replacement mounting inserts, as applicable, shall be examined to determine compliance with the requirements of this specification. Examination shall include, but shall not be limited to, the defects listed in table III, as applicable.

TABLE III. Classification of defects.

Category	Defect
Critical	None defined.
Major A:	
101	Type not as specified.
102	Style and design not as specified.
103	Components missing (quantity not as specified).
104	Mating components do not fit properly.
105	Upper member does not move freely on condylar elements (shouldered pins).
106	Upper member does not return to centric position by spring action when released from any eccentric position.
107	Upper member not removable without removal of spring.
108	Casting not free from cracks or other defects which detract from its appearance or affect its serviceability.

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TABLE III. Classification of defects. - Continued

Category	Defect
Major B:	
151	Dimensions not as specified.
Minor:	
201	Finish not as specified.
202	Plating not free of blisters, or other defects that may cause peeling.
203	Knurling not as specified, missing or incomplete.
204	Articulator not permanently and legibly marked as specified.
205	Service data missing.
206	Spare parts missing.

4.2.3 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table IV. The unit of product for testing purposes shall be one complete articulator (including mounting inserts) or one package of replacement mounting inserts, as applicable.

TABLE IV. Test criteria.

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
	Chemical analysis	Table I	4.2.4.1	4.2.3.1	*
Spring	Plating thickness	Table I	4.2.4.2	S-2	*
Spring	Hardness	Table I and 3.2	4.2.4.3	S-2	*
Articulator	Weight	3.5	4.2.4.4	S-2	4.0

* Acceptance number shall be zero for all sample sizes.

4.2.3.1 Raw materials. A sufficient quantity from each lot of each metal used in the construction of the articulators or replacement mounting inserts (as applicable) shall be selected for chemical analysis in accordance with 4.2.4.1.

4.2.4 Test procedures. Tests shall be conducted to determine compliance with the requirements of this specification.

4.2.4.1 Chemical analysis of the metals. In the absence of certificates of quality for the raw material, chemical analysis shall be conducted in accordance with Method 111 or 112 of FED-STD-151.

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4.2.4.2 Plating thickness. Thickness of plating measurements shall be made by any of the test methods described in QQ-N-290.

4.2.4.3 Hardness. Hardness shall be determined in accordance with American Society for Testing and Materials (ASTM) Designation E 18 (Rockwell) or E 92 (Vickers or Diamond Pyramid).

4.2.4.4 Weight. The articulator shall be weighed on a calibrated scale of suitable sensitivity and accuracy.

4.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. Each articulator, type I, or one package containing twelve replacement mounting inserts, type II, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit packaging.

5.1.2.1.1 Type I. Each articulator, complete, along with 12 replacement mounting inserts, shall be clean, dry, and packaged in accordance with MIL-P-116, method III, as follows. The 12 replacement mounting inserts shall be packaged in a clear plastic box having a hinged cover and friction closure. The complete articulator shall be wrapped in tissue paper and packaged along with the replacement mounting inserts in a box conforming to PPP-B-676, type I. Closure shall be as specified in the box specification.

5.1.2.1.2 Type II. Twelve mounting inserts shall be packaged in accordance with MIL-P-116, method III, in a clear plastic box having a hinged cover and friction type closure. The closure of the box shall be further secured with a strip of pressure sensitive cellophane tape.

5.1.2.2 Intermediate package.

5.1.2.2.1 Type I. An intermediate package shall not be required.

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5.1.2.2.2 Type II. Twelve unit packages shall be packed in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each type I articulator along with 12 replacement mounting inserts, or one package, type II, containing 12 mounting inserts shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Twenty-four unit packages containing type I articulators or twenty-four intermediate packages containing type II replacement mounting inserts shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Twenty-four unit packages containing type I articulators or twenty-four intermediate packages containing type II replacement mounting inserts shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged articulators or replacement mounting inserts shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Unit, intermediate, and exterior containers shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Type required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 National Stock Number (NSN) coverage. This specification covers the following items appearing in the Federal Supply Catalog:

NSN	Item Identification	Specification Designation
6520-01-031-3127	ARTICULATOR, DENTAL, Plain Line, Lingual Access Design	Type I
6520-01-036-7257	MOUNTING INSERT, DENTAL ARTICULATOR, Disposable, 12s	Type II

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD-MB

Preparing activity:
DLA-DM

Project No. 6520-1471

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-A-37819A ARTICULATOR, DENTAL, PLAIN LINE, LINGUAL ACCESS DESIGN, WITH CONDYLAR AND AUDITORY PINS, AND REPLACEMENT MOUNTING INSERTS.

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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